

# Requirement Analysis Document (RAD)

**Project Title:**

**Online Shopping Platform Database System (Implementation: "ShopEase")**

**Student Name:** ARCHANA

**Register Number:** ASML25002

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## 1. Executive Project Summary

ShopEase is an online shopping platform that allows customers to create accounts, browse products, add items to a shopping cart, place orders, make secure online payments, and track deliveries. Sellers can list products, manage inventory, and process customer orders, while administrators oversee the entire platform.

A Relational Database Management System (RDBMS) such as MySQL or PostgreSQL is used to maintain accurate customer, product, payment, and order information.

The database ensures ACID (Atomicity, Consistency, Isolation, Durability) properties.

### Example:

- **Atomicity:** If payment fails during checkout, the order is cancelled and inventory remains unchanged.
  - **Consistency:** The system prevents invalid orders such as purchasing out-of-stock products.
  - **Isolation:** Multiple customers purchasing the same product simultaneously cannot exceed available stock.
  - **Durability:** Once an order is confirmed, the information remains permanently stored.
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## 2. Core Business Domains & Features

### 2.1 Customer Management

This module manages customer registration, login, profile details, and addresses.

#### Business Rules

- Every customer must register using a unique email address.
- Each customer receives a unique Customer ID.
- Customers can update personal information and delivery addresses.

- A customer may browse products without placing any orders.
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## **2.2 Product & Category Management**

This module stores product information, categories, prices, brands, and stock.

### **Business Rules**

- Every product belongs to one category.
  - Every product has a unique Product ID.
  - Product price must always be greater than zero.
  - Stock quantity cannot be negative.
  - Sellers can update product information and stock availability.
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## **2.3 Shopping Cart & Order Management**

This module manages shopping carts and customer orders.

### **Business Rules**

- Customers can add multiple products to their cart.
- One order may contain multiple products.
- Each order has a unique Order ID.
- Order status follows:

**Cart → Placed → Paid → Packed → Shipped → Delivered → Cancelled**

- Customers cannot order products that are out of stock.
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## **2.4 Payment & Shipment Management**

This module handles payment processing and order delivery.

### **Business Rules**

- Every order has one payment record.
- Payment methods include:
  - UPI
  - Credit Card
  - Debit Card

- Net Banking
    - Cash on Delivery
  - Shipment begins only after successful payment (except Cash on Delivery).
  - Each shipment has a unique Tracking ID.
  - Customers can track their orders until delivery.
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### **3. Actor Profiles & System Interactions**

#### **1. Customer**

##### **Can Read**

- Product details
- Categories
- Prices
- Reviews
- Order history
- Shipment status

##### **Can Write/Edit**

- Personal profile
- Shopping cart
- Orders
- Reviews

##### **Safety Boundary**

Customers cannot access other customers' information or modify product details.

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#### **2. Seller**

##### **Can Read**

- Product inventory
- Customer orders
- Sales reports

##### **Can Write/Edit**

- Add products
- Update prices
- Update stock
- Process orders

### **Safety Boundary**

Sellers cannot access customer payment details or modify other sellers' products.

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## **3. Delivery Partner**

### **Can Read**

- Delivery address
- Order details
- Shipment status

### **Can Write/Edit**

- Update shipment status
- Mark orders as delivered

### **Safety Boundary**

Delivery staff cannot edit product information or payment records.

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## **4. System Administrator**

### **Can Read/Write**

- Complete access to all database tables.
  - Manage users, sellers, products, orders, and reports.
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## **4. Analytical Reporting Requirements**

### **Report 1: Best Selling Products**

#### **Business Value**

Identifies products with the highest sales.

#### **Report Includes**

- Product Name

- Total Quantity Sold
  - Revenue Generated
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## **Report 2: Customer Purchase Report**

### **Business Value**

Finds customers who spend the most.

### **Report Includes**

- Customer Name
  - Email
  - Number of Orders
  - Total Amount Spent
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## **Report 3: Monthly Sales Report**

### **Business Value**

Shows monthly business performance.

### **Report Includes**

- Month
  - Total Orders
  - Revenue
  - Most Popular Category
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## **5. Non-Functional Requirements & Database Constraints**

### **Data Constraints**

- Product price must be positive.
- Stock quantity cannot be negative.
- Email addresses must be unique.
- Payment amount must equal the total order amount.

### **Security**

- Customer passwords must be encrypted.

- Secure authentication is required for all users.

### **Performance**

- Product searches should return results within a few seconds.
- The system should support thousands of simultaneous users.

### **Concurrency**

- If two customers purchase the last available product simultaneously, only one order should succeed, while the other customer is informed that the item is out of stock.

### **Backup & Recovery**

- Daily database backups should be maintained.
- Order history should never be permanently deleted; cancelled orders remain stored for auditing purposes.